

☐ **5 V's of Big Data** : Volume, Velocity, Variety, Veracity, Value

☐ Applying the 5 V's to real-world cases

Descriptive Statistics

☐ Stem-and-leaf plots

☐ Mean, Median, Mode

☐ Variance, Standard Deviation

☐ Sample Skewness (symmetry)

☐ Sample Kurtosis (peakedness)

☐ Moments about the mean (1st to 4th)

☐ Percentile calculation (e.g., 10th percentile)

☐ Quartiles (Q1, Q3)

☐ Interpreting all of the above statistical summaries

☐ **Basic linear algebra**

☐ Steps to compute LU factorization of a matrix

☐ Contingency table construction and formula for Chi-square test

Linear Regression

☐ Concept of dependent and independent variables

☐ Least Squares Method

☐ Calculating slope and intercept

☐ Making predictions using the regression equation

☐ Understanding residuals and Sum of Squared Errors (SSE)

☐ Linear regression formula

☐ Total sum of squares (TSS)

☐ Residual sum of squares (RSS)

☐ R^2 interpretation (Goodness of fit)

☐ MAE (Mean Absolute Error)

☐ MSE (Mean Squared Error)

☐ Error interpretation in predictions

☐ **Concept of clustering**

☐ Steps in K-Means:

- Initialize centroids
- Assign clusters using Euclidean distance

- Recalculate centroids
- Repeat

☐ Distance formula

☐ Interpreting the results of clusters

☐ **ETL = Extract, Transform, Load**

☐ Identifying Data sources in given case study

☐ Transformation steps (for example cleaning, normalization, integration)

☐ Loading into data warehouse

☐ Types of Data:

- Nominal
- Ordinal
- Discrete
- Continuous
- Boolean/Binary

☐ Encoding Techniques:

- Label Encoding (for binary/ordinal)
- One-Hot Encoding (for nominal/categorical)
- No encoding needed for numerical values
- Considerations for ML-readiness

Data Preprocessing

☐ Handling missing values

☐ Removing duplicate records

☐ Identifying and managing outliers

☐ Encoding categorical variables (Label Encoding, One-Hot Encoding)

☐ Feature scaling (Standardization, Normalization)

☐ Splitting data into training and testing sets

☐ Supervised Learning

☐ Unsupervised Learning

☐ Reinforcement Learning

☐ Semi-supervised Learning

- ☐ Real-world examples for each type
- ☐ Identifying ML types from problem scenarios

Naive Bayes Classifier

- ☐ Basic probability concepts
- ☐ Prior and likelihood
- ☐ Naive Bayes formula
- ☐ Binary classification with Yes/No features
- ☐ Laplace smoothing (optional)
- ☐ Choosing class with highest posterior probability

Data Visualization

- ☐ Gestalt Principles:
 - ☐ Similarity
 - ☐ Proximity
 - ☐ Closure
 - ☐ Continuity
 - ☐ Common Fate
 - ☐ Figure/Ground
- ☐ Apply them to charts (scatter plot, pie chart, clustered bar, etc.) / dashboards